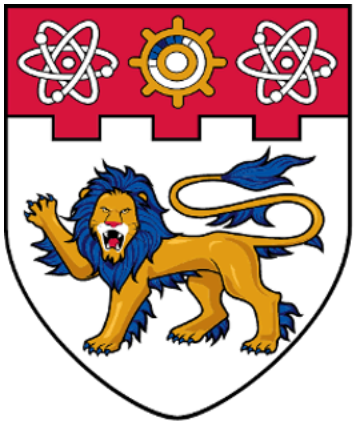


NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

SC2006: SOFTWARE ENGINEERING

Lab 2 Deliverables

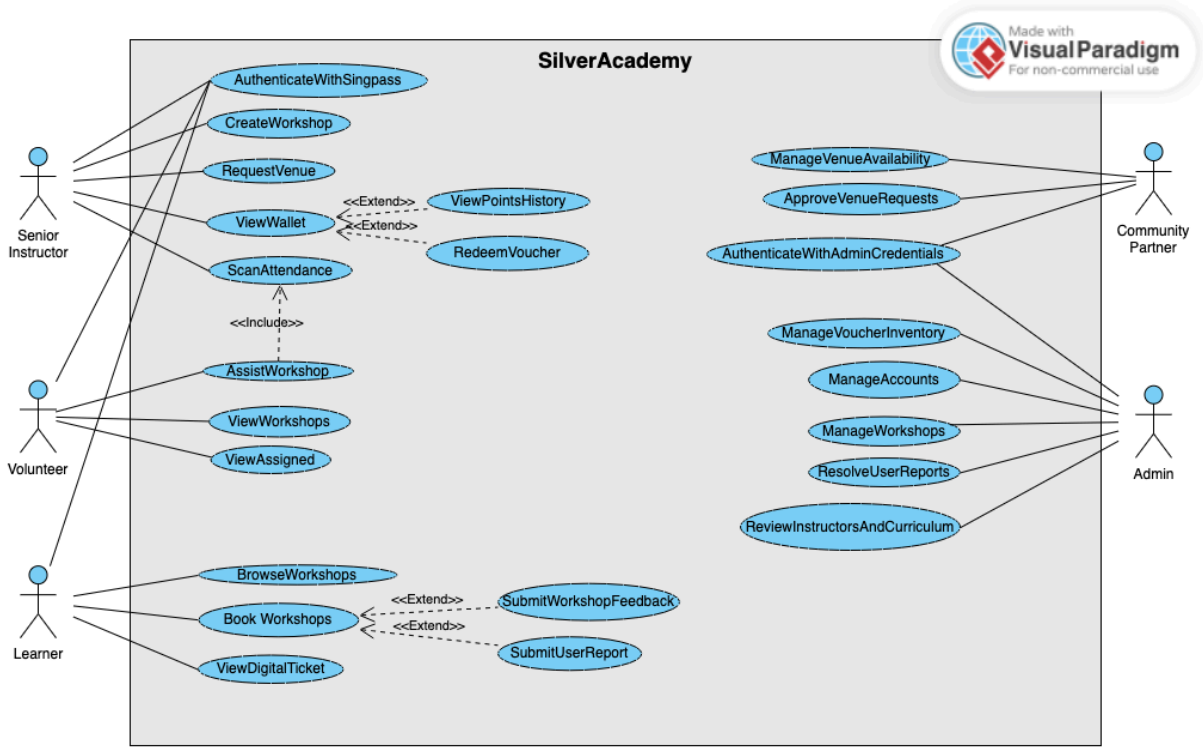
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1. Use Case Diagram



Link: https://online.visual-paradigm.com/share.jsp?id=3437303033383

2. Functional Requirements

1. Global Prerequisites

1. SilverAcademy shall provide a single registration and login point using **Singpass** for Learners, Senior Instructors and Volunteers.
2. SilverAcademy shall verify the user's age through the authenticated Singpass profile before enabling Senior Instructor functionality.
3. For an authenticated user verified as **50 years old or above**, SilverAcademy shall allow the user to switch between **Senior Instructor Mode** and **Learner Mode** without logging out.
4. SilverAcademy shall display only the functions permitted for the user's current role and account status.

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2. Senior Instructor

1. SilverAcademy shall verify that a Senior Instructor has an **approved Instructor profile** before enabling workshop creation functions.
2. SilverAcademy shall allow Senior Instructors to enter a workshop title, description, maximum capacity and required materials.
3. SilverAcademy shall validate all mandatory workshop fields before allowing the workshop to be submitted.
4. SilverAcademy shall allow Senior Instructors to select an available neighbourhood venue using an interactive map.
5. SilverAcademy shall display the available dates and time slots for the selected venue.
6. SilverAcademy shall allow Senior Instructors to submit a venue booking request for a selected workshop, venue, date and time.
7. SilverAcademy shall provide Senior Instructors with a QR scanning interface for verifying Learner attendance.
8. SilverAcademy shall verify that a scanned QR code belongs to a registered Learner for the active workshop session.
9. SilverAcademy shall update a Learner's attendance status to **Attended** after a successful QR verification.
10. SilverAcademy shall award the corresponding points to the Senior Instructor after a successful attendance verification.

11. SilverAcademy shall allow Senior Instructors to view their current points balance and available voucher rewards.
12. SilverAcademy shall allow Senior Instructors to view a chronological history of points earned and redeemed.
13. SilverAcademy shall allow an eligible Senior Instructor to select an available voucher denomination and submit a redemption request.
14. SilverAcademy shall assign one available voucher code to the Senior Instructor after a successful redemption.
15. SilverAcademy shall deduct the required points from the Senior Instructor's wallet after successfully allocating the voucher.

---

3. Learner

1. SilverAcademy shall allow authenticated Learners to view available workshops.
2. SilverAcademy shall allow Learners to search workshops using keywords.
3. SilverAcademy shall allow Learners to filter workshops by **category, date and location**.This is much more verifiable than: provide a search and filter system. You can literally test: Search = Cooking → only matching workshops appear. Date = 20–25 September → only workshops within that range appear.
4. SilverAcademy shall allow Learners to sort available workshops by distance from their selected or permitted location.
5. SilverAcademy shall display recommended workshops based on workshop popularity and/or proximity.
6. SilverAcademy shall allow Learners to view the selected workshop's title, description, venue, date, time, capacity and material requirements.
7. SilverAcademy shall allow Learners to book an available workshop with at least one remaining slot.
8. SilverAcademy shall prevent a Learner from booking the same workshop more than once.
9. SilverAcademy shall prevent the number of confirmed bookings from exceeding the workshop's maximum capacity.
10. SilverAcademy shall display a booking confirmation after a successful workshop registration.
11. SilverAcademy shall generate a unique digital ticket containing a QR code for every confirmed workshop booking.
12. SilverAcademy shall allow Learners to access their active digital tickets from their booking history.

13. SilverAcademy shall allow Learners who have attended a workshop to submit a rating and written feedback.
14. SilverAcademy shall prevent Learners who have not attended the workshop from submitting workshop feedback.
15. SilverAcademy shall allow Learners to submit a report regarding a workshop, user behaviour, venue safety issue or software issue.
16. SilverAcademy shall generate a unique reference number for each submitted report.

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4. Volunteer

1. SilverAcademy shall allow authenticated Volunteers to view workshops requiring assistance.
2. SilverAcademy shall allow Volunteers to view available assistance opportunities based on location.
3. SilverAcademy shall allow Volunteers to opt in to assist with an available workshop.
4. SilverAcademy shall prevent a Volunteer from opting in to the same workshop more than once.
5. SilverAcademy shall verify that a Volunteer is assigned to a workshop before granting access to its Learner class list.
6. SilverAcademy shall allow assigned Volunteers to view the workshop details and registered Learner list.
7. SilverAcademy shall allow assigned Volunteers to scan Learner QR codes for attendance verification.
8. SilverAcademy shall update the Learner's attendance record after a valid QR code is scanned by an assigned Volunteer.

---

5. Community Partner

1. SilverAcademy shall allow authorised Community Partners to authenticate through the management portal using their User ID and password.
2. SilverAcademy shall allow Community Partners to view facilities registered under their organisation.
3. SilverAcademy shall allow Community Partners to mark facility time slots as **Available, Unavailable or Under Maintenance**.

4. SilverAcademy shall prevent a Community Partner from making an approved workshop booking slot unavailable without resolving the existing booking.
5. SilverAcademy shall allow Community Partners to view pending venue requests for their facilities.
6. SilverAcademy shall allow Community Partners to approve a pending venue request.
7. SilverAcademy shall allow Community Partners to reject a pending venue request and optionally provide a reason.
8. SilverAcademy shall mark an approved venue time slot as **Booked**.
9. SilverAcademy shall notify the relevant Senior Instructor when a venue request is approved or rejected.

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6. Admin

1. SilverAcademy shall allow authorised Admins to authenticate through the management portal using enterprise credentials.
2. SilverAcademy shall allow Admins to search registered user accounts.
3. SilverAcademy shall allow Admins to view the role, account status and relevant profile information of a selected user.
4. SilverAcademy shall allow Admins to flag a user account for monitoring.
5. SilverAcademy shall allow authorised Admins to suspend a user account.
6. SilverAcademy shall allow authorised Admins to reactivate a suspended account.
7. SilverAcademy shall record all account status changes in the administrative audit log.

Workshop Management

8. SilverAcademy shall allow Admins to search and filter workshop listings.
9. SilverAcademy shall allow Admins to edit permitted workshop information.
10. SilverAcademy shall allow Admins to archive a workshop from public discovery.
11. SilverAcademy shall allow authorised Admins to change a workshop's status according to predefined administrative rules.
12. SilverAcademy shall preserve existing booking and attendance records when a workshop is archived.
13. SilverAcademy shall create an incident record when a valid user report is submitted.
14. SilverAcademy shall add each new incident record to the central Admin Incident Queue.

15. SilverAcademy shall allow Admins to view the details and supporting evidence of an incident.
16. SilverAcademy shall allow Admins to change an incident status to **Under Review, Awaiting Information, Resolved or Closed**.
17. SilverAcademy shall allow Admins to request additional information from the relevant user.
18. SilverAcademy shall allow Admins to enter a resolution for an incident.
19. SilverAcademy shall allow authorised Admins to flag or suspend a user account when required during incident resolution.
20. SilverAcademy shall allow authorised Admins to cancel a workshop when required during incident resolution.
21. SilverAcademy shall notify relevant users when an incident resolution results in an action affecting them.
22. SilverAcademy shall record incident status changes, administrative actions and resolutions in the audit log.

Voucher Management

23. SilverAcademy shall allow Admins to upload pre-purchased voucher codes into the voucher inventory.
24. SilverAcademy shall validate the format of uploaded voucher codes before storing them.
25. SilverAcademy shall reject voucher codes that already exist in the voucher inventory.
26. SilverAcademy shall maintain the status of each voucher as **Available, Claimed, Invalid or Removed**.
27. SilverAcademy shall allow Admins to view voucher inventory grouped by provider, denomination and status.
28. SilverAcademy shall mark a voucher as **Claimed** and associate it with the Senior Instructor's account when successfully redeemed.
29. SilverAcademy shall record voucher uploads, status changes and redemptions in the administrative audit log.

Instructor and Curriculum Approval

30. SilverAcademy shall allow Admins to view Senior Instructor profiles awaiting approval.
31. SilverAcademy shall allow Admins to review the submitted information of a Senior Instructor.
32. SilverAcademy shall allow Admins to approve an eligible Senior Instructor profile.

33. SilverAcademy shall allow Admins to reject a Senior Instructor profile with a recorded reason.
34. SilverAcademy shall allow Admins to review submitted workshop curriculum.
35. SilverAcademy shall allow Admins to approve a curriculum that satisfies the defined safety requirements.
36. SilverAcademy shall allow Admins to return a curriculum for revision with a recorded reason.

---

3. Non-Functional Requirements

1. Security
    1. Identity verification via Singpass shall be mandatory for every Senior Instructor and Learner accessing protected SilverAcademy functions.
    2. SilverAcademy shall enforce role-based access control for Senior Instructors, Learners, Volunteers, Community Partners and Admins.
    3. Administrative and Community Partner passwords shall not be stored in plaintext and shall be protected using a one-way cryptographic hashing mechanism.
    4. SilverAcademy shall record successful and failed administrative login attempts and security-sensitive administrative actions, including the actor and timestamp.
2. Usability
    1. The user interface for Senior Instructors shall provide adjustable text sizing and oversized touch controls.
    2. Primary interactive controls shall have a minimum touch target of **44 × 44 pixels**.
    3. The main Senior Instructor dashboard shall provide direct access to **Create Workshop, Scan Attendance and View Wallet**.
    4. SilverAcademy shall display clear error messages when users enter invalid or incomplete information.
3. Performance
    1. Selecting **“Scan Attendance”** shall launch the camera view within **3 seconds** under normal device and network conditions.
    2. SilverAcademy shall validate a valid Learner QR code and complete attendance detection within **3 seconds** under normal operating conditions.
    3. Selecting **“Redeem Voucher”** shall process the request and deduct wallet credits within **5 seconds** under normal operating conditions.
    4. SilverAcademy shall return workshop search results within **2 seconds** for the expected project dataset.
    5. Standard application requests shall return a response within **2 seconds**, excluding delays caused by external services such as Singpass and OneMap.
4. Reliability
    1. SilverAcademy shall ensure that a workshop booking either completes successfully or does not modify the booking record.
    2. Concurrent booking requests shall not result in confirmed bookings exceeding the workshop capacity.
    3. A Learner shall not be marked as attended more than once for the same workshop session.
    4. Voucher redemption shall either complete both wallet credit deduction and voucher allocation or perform neither operation.
    5. A confirmed digital Learner ticket shall remain accessible on the device when network connectivity is temporarily unavailable.
    6. If an external service becomes temporarily unavailable, SilverAcademy shall display an appropriate error message without corrupting existing application data.

5. Scalability
    1. SilverAcademy shall support up to **10,000 concurrent users** while maintaining core application functionality.
    2. The platform shall support up to **10,000 simultaneous OneMap-related requests** without causing the application server to fail.
    3. Under the defined concurrent-user load, at least **95% of standard application requests** shall complete within **2 seconds**, excluding delays caused by external APIs.
6. Maintainability
    1. SilverAcademy shall separate major application components into modules for authentication, workshops, bookings, attendance, vouchers and user management.
    2. External services such as Singpass and OneMap shall be accessed through defined **API/SDK interfaces** to reduce dependency on specific implementations.
    3. Voucher inventory data shall be stored in a dedicated database or logical data store isolated from sensitive user information.
    4. Documentation shall be maintained for major APIs, database entities and external SDK/API integrations.
7. Platform and Compatibility
    1. The SilverAcademy **mobile application** shall support the primary workflows of Senior Instructors, Learners and Volunteers.
    2. The SilverAcademy **web application** shall support the management workflows of Admins and Community Partners.
    3. The mobile application shall use a supported **camera/QR scanning SDK or interface** for attendance verification.
    4. SilverAcademy shall integrate with the **Singpass authentication service** and **OneMap API** through their supported APIs or SDKs.
    5. The Admin and Community Partner web application shall remain usable on supported desktop and tablet screen sizes without loss of core functionality.

4. Use Case Descriptions

I. Universal Requirements

A. AuthenticateWithSingpass

Use Case ID:	#0-1
Use Case Name:	AuthenticateWithSingpass

Actor(s):	Senior Instructor, Learner, Volunteer
Description:	Allows a user to authenticate their identity securely through the national Singpass service before accessing their respective app dashboards.
Preconditions:	1. The user has a valid Singpass account. 2. SilverAcademy is available.
Postconditions:	The user is successfully authenticated, and if verified as 50+ years of age, the application unlocks the dual-role Instructor/Learner toggle interface.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects **"Login with Singpass"**. 2. SilverAcademy redirects the user to the Singpass authentication service. 3. The user completes Singpass authentication. 4. Singpass returns the authentication result to SilverAcademy. 5. SilverAcademy verifies the authentication result. 6. SilverAcademy retrieves the user's profile and authorised role. 7. SilverAcademy creates an authenticated session. 8. If the user is eligible for both Senior Instructor and Learner access, SilverAcademy enables the role toggle. 9. SilverAcademy displays the appropriate interface.
Alternative Flows:	None

Exceptions:	1. Ex1: If Singpass authentication fails, SilverAcademy shall reject the login attempt and display an error message. 2. Ex2: If the Singpass service is unavailable, SilverAcademy shall display an appropriate service unavailable message. 3. Ex3: If the user's profile or role cannot be retrieved, SilverAcademy shall deny access to protected functions and display an error message.
Includes:	None
Special Requirements:	NFR 7.4
Assumptions:	The user owns a mobile device equipped with working biometric components or a configured Singpass passcode sequence.
Notes and Issues:	None

B. AuthenticateWithAdminCredentials

Use Case ID:	#0-2
Use Case Name:	AuthenticateWithAdminCredentials

Actor(s):	Admin, Community Partner
Description:	Allows administrative platform operators to securely log into their respective management dashboards using a User ID and password credential interface.
Preconditions:	1. The Admin or Community Partner account has been securely pre-created and

	authorized by the core system backend. 2. The internal corporate authentication service is online and active.
Postconditions:	The administrative user is successfully authenticated and granted access to their designated back-office dashboards.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The administrative actor navigates to the dedicated SilverAcademy Management portal. 2. SilverAcademy renders the login page containing User ID and Password text input fields. 3. The actor inputs their designated administrative alphanumeric username and hidden password. 4. The actor clicks the "Secure Login" action command button. 5. SilverAcademy safely hashes the input password string and cross-references it against the encrypted records inside the isolated user data registry. 6. The system identifies the user's role permission flags (Admin or Community Partner). 7. SilverAcademy logs the user in and opens their specialized back-office workflow console.
Alternative Flows:	None

Exceptions:	1. Ex1: If the inputted User ID does not exist or the password string fails the cryptographic hash match, SilverAcademy blocks system entry, clears the form inputs, and displays a red "Invalid Credentials. Access Denied" security notification. 2. Ex2: If an account is flagged by the system as "Suspended" or "Inactive", SilverAcademy halts the login workflow and displays a message directing the user to contact the platform team.
Includes:	None
Special Requirements:	None
Assumptions:	Administrative users maintain robust password complexity standards to safeguard public data platforms.
Notes and Issues:	All successful logins and failed access attempts must generate an automated entry inside the system audit log repository for governance tracking.

II. For Functional Requirement #1 - Senior Instructor

A. CreateWorkshop

Use Case ID:	#1-2
Use Case Name:	CreateWorkshop

Actor(s):	Senior Instructor
Description:	Allows a Senior Instructor to create a new workshop by entering the required workshop information.
Preconditions:	1. The Senior Instructor is logged in and authenticated. 2. The Senior Instructor has an approved profile.
Postconditions:	A new workshop containing the submitted information is created and stored in SilverAcademy.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Senior Instructor selects **"Create Workshop"**. 2. SilverAcademy verifies that the Senior Instructor is authorised to create workshops. 3. The Senior Instructor enters the workshop title, description, maximum capacity and material requirements. 4. SilverAcademy validates the entered workshop information. 5. SilverAcademy creates the workshop with the appropriate registration status.

	6. SilverAcademy displays the created workshop to the Senior Instructor.
Alternative Flows:	None
Exceptions:	1. Ex1: If any required workshop information is missing or invalid, SilverAcademy shall display an error message and request correction before creating the workshop.
Includes:	None
Special Requirements:	The interface must adhere to SS 618 senior-centric UX standards (large input fields and text targets).
Assumptions:	None
Notes and Issues:	None

B. RequestVenue

Use Case ID:	#1-3
Use Case Name:	RequestVenue

Actor(s):	Senior Instructor
Description:	Allows a Senior Instructor to browse available venues using a geospatial map and request a specific location for their workshop.
Preconditions:	1. The Senior Instructor is logged in and authenticated. 2. A workshop proposal has been created.
Postconditions:	A venue booking request is successfully tied to the workshop and sent to the respective Community Partner for approval.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Senior Instructor selects **"Request Venue"** for a workshop. 2. SilverAcademy retrieves available venues and displays them on the map. 3. The Senior Instructor selects a preferred venue. 4. The Senior Instructor selects an available date and time slot.

	5. SilverAcademy checks whether the selected venue and time slot are available. 6. SilverAcademy submits the venue request to the Community Partner. 7. SilverAcademy displays the request status to the Senior Instructor.
Alternative Flows:	None
Exceptions:	1. Ex1: If the selected venue or time slot is unavailable, SilverAcademy shall inform the Senior Instructor and request another selection. 2. Ex2: If the map service is unavailable, SilverAcademy shall provide the available venues using a list-based view.
Includes:	None
Special Requirements:	NFR 7.4
Assumptions:	Community Partners keep their physical facility operating hours and block-out dates updated in the system.
Notes and Issues:	None

C. ScanAttendance

Use Case ID:	#1-4
Use Case Name:	ScanAttendance

Actor(s):	Senior Instructor
Description:	Allows a Senior Instructor to use their smartphone camera to scan a Learner's digital ticket QR code to verify attendance.
Preconditions:	1. The Senior Instructor is logged in and authenticated. 2. The scheduled workshop date and time are active.
Postconditions:	The Learner's attendance status is updated to "Attended" and a point is awarded to the Senior Instructor’s wallet.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The authorised Senior Instructor or Volunteer selects **"Scan Attendance"**. 2. SilverAcademy verifies that the user is authorised to record attendance for the selected workshop. 3. SilverAcademy launches the device camera. 4. The user scans the Learner's Digital Ticket

	QR code. 5. SilverAcademy validates the QR code. 6. SilverAcademy verifies that the Learner has a valid booking for the workshop. 7. SilverAcademy records the Learner as attended. 8. If the scanner is operated by a Senior Instructor, SilverAcademy awards the applicable instructor point. 9. SilverAcademy displays the attendance confirmation.
Alternative Flows:	Volunteer Assistant Mode: If the Senior Instructor requests help, the Volunteer logs into their own dashboard, activates their assistant scanner camera, and scans the Learner's QR code to fulfill steps 3–6 on behalf of the senior.
Exceptions:	1. Ex1: If the QR code is invalid, SilverAcademy shall reject the scan and display an error message. 2. Ex2: If the Learner does not have a valid booking for the workshop, SilverAcademy shall reject the attendance record. 3. Ex3: If the Learner has already been marked as attended, SilverAcademy shall reject the duplicate attendance attempt. 4. Ex4: If camera scanning is unavailable, SilverAcademy shall allow the authorised user to enter the Learner's booking ID manually.
Includes:	None
Special Requirements:	NFR 7.3

Assumptions:	The Learner has a Digital Ticket (QR code).
Notes and Issues:	None

D. ViewWallet

Use Case ID:	#1-5
Use Case Name:	ViewWallet

Actor(s):	Senior Instructor
Description:	Allows a Senior Instructor to review their workshop completion logs, check their wallet credit balance, and view available reward tiers.
Preconditions:	The Senior Instructor is logged in and authenticated.
Postconditions:	The Senior Instructor's current wallet balance and available voucher information are displayed
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Senior Instructor selects the "View Wallet" option from their primary control panel. 2. The application queries the isolated balance database table for the user's instructorID. 3. SilverAcademy calculates and displays the current unspent milestone total and total historical lifetime earnings. 4. The system populates a list of selectable supermarket voucher redemption cards below the balance display.
Alternative Flows:	View Milestone History: The instructor can click the

	"View Points History" text link, which triggers an optional data extension to populate an itemized list of historical sessions.
Exceptions:	Ex1: If the database connection drops, SilverAcademy renders a local cache error: "Unable to retrieve current balance. Trying to reconnect."
Includes:	None
Special Requirements:	Layout typography must scale to oversized options in compliance with SS 618 usability parameters
Assumptions:	The wallet balance accurately reflects the historical count of verified attendance scans.
Notes and Issues:	None

E. ViewPointsHistory

Use Case ID:	#1-6
Use Case Name:	ViewPointsHistory

Actor(s):	Senior Instructor
Description:	Provides an itemized log of historical workshops conducted and exact timestamps of when wallet credits were added or voucher claims occurred.
Preconditions:	The Senior Instructor is currently active inside the ViewWallet base use case dashboard screen (#1-6).
Postconditions:	The historical activity sub-menu expands to render a chronological ledger feed.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The Senior Instructor clicks on the "View Points History" command element. 2. SilverAcademy triggers an extension point branching from the primary wallet panel view. 3. The backend fetches transaction row records associated with the target instructorID ledger. 4. The screen populates a clean list showing dates, workshop titles, check-in totals, and voucher deductions.

Alternative Flows:	None
Exceptions:	Ex1: If the database connection drops, SilverAcademy renders a local cache error: "Unable to retrieve current balance. Trying to reconnect."
Includes:	None
Special Requirements:	Extended lists must support simple, smooth touch scrolling optimized for elderly accessibility.
Assumptions:	Ledger logs are read-only metrics and cannot be altered by the user profile.
Notes and Issues:	None

F. RedeemVoucher

Use Case ID:	#1-7
Use Case Name:	RedeemVoucher

Actor(s):	Senior Instructor
Description:	Allows a Senior Instructor to exchange their accumulated wallet balance for a supermarket grocery voucher code.
Preconditions:	1. The Senior Instructor is logged in and authenticated. 2. The Senior Instructor's wallet balance satisfies the minimum workshop milestone requirement to unlock a voucher tier.
Postconditions:	The Senior Instructor's wallet balance is deducted, and a unique voucher barcode is assigned to their profile and displayed on-screen.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The Senior Instructor selects **"Redeem Voucher"**. 2. SilverAcademy displays the available voucher types and required points. 3. The Senior Instructor selects a voucher. 4. SilverAcademy verifies that the Senior Instructor has sufficient wallet credits. 5. SilverAcademy verifies that an unclaimed voucher is available.

	6. SilverAcademy deducts the required wallet credits. 7. SilverAcademy marks the selected voucher as claimed and associates it with the Senior Instructor. 8. SilverAcademy displays the voucher code or barcode to the Senior Instructor.
Alternative Flows:	None
Exceptions:	1. Ex1: If the Senior Instructor has insufficient wallet credits, SilverAcademy shall reject the redemption and display the required points. 2. Ex2: If no unclaimed voucher is available, SilverAcademy shall reject the redemption without deducting wallet credits. 3. Ex3: If the redemption transaction fails, SilverAcademy shall not deduct wallet credits or claim the voucher.
Includes:	None
Special Requirements:	None
Assumptions:	There are vouchers stocked in the internal database.
Notes and Issues:	None

III. For Functional Requirement #2 - Learner

A. BrowseWorkshops

Use Case ID:	#2-1
Use Case Name:	BrowseWorkshops

Actor(s):	Learner
Description:	Allows a Learner to find workshops using keywords, geospatial proximity, categories, and date filters.
Preconditions:	The Learner is logged into SilverAcademy.
Postconditions:	SilverAcademy returns a curated list of workshops matching the specified search and filter criteria.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The Learner navigates to the **Workshop Directory**. 2. SilverAcademy displays available workshops. 3. The Learner enters a keyword or selects one or more search filters. 4. The Learner may filter workshops by **location, category and date**. 5. SilverAcademy retrieves workshops matching all selected search criteria. 6. SilverAcademy displays the matching workshops. 7. The Learner selects a workshop to view its

	details.
Alternative Flows:	The Learner resets active filters, prompting SilverAcademy to restore the default recommended workshop list.
Exceptions:	1. Ex1: If no workshops match the selected criteria, SilverAcademy shall display a **"No workshops found"** message. 2. Ex2: If location access is unavailable, SilverAcademy shall allow the Learner to browse workshops without proximity-based filtering. 3. Ex3: If the workshop database cannot be reached, SilverAcademy shall display an appropriate error message.
Includes:	None
Special Requirements:	NFR 3.4
Assumptions:	Published workshops are active, verified, and mapped to valid coordinates.
Notes and Issues:	Proximity search requires device location access permission.

B. BookWorkshops

Use Case ID:	#2-2
Use Case Name:	BookWorkshops

Actor(s):	Learner
Description:	Allows a Learner to register and secure a vacant slot for an upcoming workshop session.
Preconditions:	1. The Learner is authenticated. 2. The workshop status is "Open for Registration" with at least one vacant seat.
Postconditions:	The workshop's remaining seat count decreases by 1, the Learner's profile is added to the roster, and a Digital Ticket is issued.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Learner navigates to the selected workshop's details page. 2. The Learner selects **"Book Workshop"**. 3. SilverAcademy verifies that the workshop is open for registration. 4. SilverAcademy verifies that at least one seat is available. 5. SilverAcademy verifies that the Learner has not already registered for the workshop. 6. SilverAcademy registers the Learner for the workshop.

	7. SilverAcademy decreases the workshop's available seat count by 1. 8. SilverAcademy generates a unique Digital Ticket containing a QR code. 9. SilverAcademy displays the booking confirmation and Digital Ticket.
Alternative Flows:	None
Exceptions:	1. Ex1: If the session reaches maximum capacity before confirmation is processed, SilverAcademy alerts the Learner that the session is full and does not charge/book. 2. Ex2: If the Learner has already registered for this session, SilverAcademy blocks the request and displays a duplicate booking alert.
Includes:	None
Special Requirements:	NFR 4.2
Assumptions:	All workshops listed under this program are free admission for participants.
Notes and Issues:	Send an automated push or in-app confirmation upon successful booking.

C. ViewDigitalTicket

Use Case ID:	#2-3
Use Case Name:	ViewDigitalTicket

Actor(s):	Learner
Description:	Allows a Learner to display their unique QR code ticket for verification upon arrival at the workshop venue.
Preconditions:	The Learner has an active, confirmed booking for an upcoming workshop.
Postconditions:	The digital pass QR code is rendered on-screen ready for optical scanning.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Learner navigates to **"My Bookings"**. 2. SilverAcademy displays the Learner's confirmed bookings. 3. The Learner selects the relevant workshop. 4. SilverAcademy retrieves the corresponding Digital Ticket. 5. SilverAcademy displays the Digital Ticket and QR code. 6. The Learner presents the QR code to an authorised Senior Instructor or Volunteer.

Alternative Flows:	None
Exceptions:	1. Ex1: If the Learner does not have a confirmed booking, SilverAcademy shall not display a valid Digital Ticket. 2. Ex2: If the workshop has concluded, SilverAcademy shall display the ticket with its attendance status. 3. Ex3: If the ticket cannot be retrieved, SilverAcademy shall display an error message and allow the Learner to retry.
Includes:	None
Special Requirements:	Ticket screen must maintain high contrast for reliable camera scanning.
Assumptions:	None
Notes and Issues:	Ticket payload should support offline display in case of poor venue connectivity.

D. SubmitWorkshopFeedback

Use Case ID:	#2-4
Use Case Name:	SubmitWorkshopFeedback

Actor(s):	Learner
Description:	Allows a Learner to submit a rating, review, and structured feedback tags for a workshop they completed.
Preconditions:	1. The Learner is authenticated. 2. The Learner's attendance status for the target workshop is marked as "Attended".
Postconditions:	The feedback is stored and aggregated into the workshop's and Senior Instructor's overall rating score.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The Learner selects an attended workshop. 2. SilverAcademy verifies that the Learner attended the workshop. 3. SilverAcademy displays the feedback form. 4. The Learner submits a rating, review and optional feedback tags. 5. SilverAcademy validates the submitted feedback. 6. SilverAcademy stores the feedback against the workshop.

	7. SilverAcademy displays a submission confirmation.
Alternative Flows:	None
Exceptions:	1. Ex1: If the Learner has not attended the workshop, SilverAcademy shall prevent feedback submission. 2. Ex2: If the feedback submission period has expired, SilverAcademy shall prevent submission and inform the Learner.
Includes:	None
Special Requirements:	None
Assumptions:	Feedback is restricted to participants who were verified as attended.
Notes and Issues:	Define an expiry window (e.g., 7 days post-workshop) after which feedback can no longer be submitted.

E. SubmitUserReport

Use Case ID:	#2-5
Use Case Name:	SubmitUserReport

Actor(s):	Learner
Description:	Allows a Learner to file a formal complaint or incident report regarding safety hazards, user conduct, venue issues, or software bugs.
Preconditions:	The Learner is logged into SilverAcademy.
Postconditions:	A new incident ticket is generated and added to the centralized Admin Incident Queue.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The Learner selects **"Report an Issue"**. 2. SilverAcademy displays the available report categories. 3. The Learner selects a report category. 4. The Learner enters a description of the issue. 5. The Learner optionally attaches supporting evidence. 6. SilverAcademy validates the report information. 7. SilverAcademy creates an incident record. 8. SilverAcademy assigns a unique report reference number. 9. SilverAcademy displays the reference

	number to the Learner.
Alternative Flows:	None
Exceptions:	1. Ex1: If required report information is missing, SilverAcademy shall prevent submission and display an error message. 2. Ex2: If an uploaded file exceeds the permitted size, SilverAcademy shall reject the file and inform the Learner. 3. Ex3: If the report cannot be submitted, SilverAcademy shall display an error message without creating an incomplete incident record.
Includes:	None
Special Requirements:	None
Assumptions:	Platform administrators routinely review incoming tickets in the queue.
Notes and Issues:	Incident reference ID should remain trackable in the Learner's support center.

IV. For Functional Requirement #3 - Volunteer

A. ViewWorkshops

Use Case ID:	#3-1
Use Case Name:	ViewWorkshops

Actor:	Volunteer
Description:	Allows a Volunteer to view nearby workshops that require assistance.
Preconditions:	1. The Volunteer is logged in and authenticated. 2. At least one workshop requiring Volunteer assistance is available.
Postconditions:	SilverAcademy displays a list of nearby workshops that require assistance.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Volunteer opens **"Volunteer Opportunities"**. 2. SilverAcademy retrieves workshops requiring volunteer assistance. 3. If location access is available, SilverAcademy orders the workshops by proximity. 4. SilverAcademy displays the available workshops. 5. The volunteer selects a workshop to view its details.
Alternative Flows:	AF-S3: If the Volunteer chooses to view all workshops instead of nearby workshops, SilverAcademy displays all available Volunteer

	opportunities.
Exceptions:	1. Ex1: If no nearby workshops require assistance, SilverAcademy informs the Volunteer that no nearby opportunities are currently available. 2. Ex2: If location information cannot be retrieved, SilverAcademy displays available workshops without proximity-based ordering.
Includes:	None
Special Requirements:	Location information is required if nearby workshops are to be ranked by proximity.
Assumptions:	Workshop location information is available in SilverAcademy.
Notes and Issues:	None

B. AssistWorkshop

Use Case ID:	#3-2
Use Case Name:	AssistWorkshop

Actor:	Volunteer
Description:	Allows a Volunteer to opt in to assist with an available workshop.
Preconditions:	1. The Volunteer is logged in and authenticated. 2. The selected workshop is still accepting Volunteer assistance.
Postconditions:	The Volunteer is recorded as having opted in to assist with the selected workshop.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Volunteer selects a workshop requiring assistance. 2. SilverAcademy displays the workshop details. 3. Volunteer selects **"Assist Workshop"**. 4. SilverAcademy verifies that the Volunteer has not already registered to assist. 5. SilverAcademy records the Volunteer as an assisting Volunteer. 6. SilverAcademy displays a confirmation.
Alternative Flows:	If the Volunteer selects Cancel, no opt-in is recorded and SilverAcademy returns to the workshop details page.
Exceptions:	1. Ex1: If the Volunteer has already opted in to

	the workshop, SilverAcademy shall prevent duplicate registration and display the current assignment status.
Includes:	None
Special Requirements:	None
Assumptions:	The workshop is open for Volunteer assistance.
Notes and Issues:	None

C. *ViewAssigned*

Use Case ID:	#3-3
Use Case Name:	ViewAssigned

Actor:	Volunteer
Description:	Allows an assigned Volunteer to view the relevant workshop details and class list required to assist the Senior Instructor.
Preconditions:	1. The Volunteer is logged in and authenticated. 2. The Volunteer has been assigned to the selected workshop.
Postconditions:	The relevant workshop information and class list are displayed to the assigned Volunteer.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Volunteer opens Assigned Workshops. 2. SilverAcademy displays the workshops assigned to the Volunteer. 3. The Volunteer selects a workshop. 4. SilverAcademy verifies that the Volunteer is assigned to the workshop. 5. SilverAcademy displays the relevant workshop details and registered Learner class list.
Alternative Flows:	None
Exceptions:	Ex1: If the Volunteer is not assigned to the selected workshop, SilverAcademy denies access to the

	workshop details and class list.
Includes:	None
Special Requirements:	Access to the class list shall be restricted to Volunteers assigned to the workshop.
Assumptions:	The workshop has at least one assigned Volunteer.
Notes and Issues:	None

D. *ScanAttendance*

Use Case ID:	#3-4
Use Case Name:	ScanAttendance

Actor:	Volunteer
Description:	Allows an assigned Volunteer to scan a Learner's QR code to verify attendance for a workshop.
Preconditions:	1. The Volunteer is logged in and authenticated. 2. The Volunteer is assigned to the workshop. 3. The workshop is active or in progress.
Postconditions:	The Learner's attendance status is updated from Registered to Attended after a successful scan.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. Volunteer selects **"Scan Attendance"**. 2. SilverAcademy verifies that the Volunteer is assigned to the workshop. 3. SilverAcademy launches the camera. 4. Volunteer scans the Learner's Digital Ticket. 5. SilverAcademy validates the QR code. 6. SilverAcademy verifies the Learner's booking. 7. SilverAcademy records the Learner as attended. 8. SilverAcademy displays the attendance confirmation.
Alternative Flows:	None
Exceptions:	1. Ex1: If the QR code is invalid, SilverAcademy

	displays an error message and does not update attendance. 2. Ex2: If the QR code has already been used, SilverAcademy informs the Volunteer that the Learner has already checked in. 3. Ex3: If the QR code belongs to a different workshop, SilverAcademy rejects the check-in.
Includes:	assistworkshop
Special Requirements:	NFR 7.3
Assumptions:	The Learner has a valid Digital Ticket QR code for the workshop.
Notes and Issues:	None

V. For Functional Requirement #4 - Community Partner

A. ManageVenueAvailability

Use Case ID:	#4-1
Use Case Name:	ManageVenueAvailability

Actor:	Community Partner
Description:	Allows a Community Partner(CP) to log, update, and manage the availability slots of the physical neighbourhood facilities (e.g. void decks, CC function rooms, badminton court), so Senior Instructors can view accurate, up-to-date venue options when requesting a workshop venue.
Preconditions:	1. CP is logged in and authenticated. 2. The organisation profile has been verified and approved by an Admin. 3. At least one facility has been registered under the Community Partner's account.
Postconditions:	The facility's availability calendar is updated in SilverAcademy and immediately reflected for Instructors browsing venues via the map.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Community Partner selects **"Manage Venue Availability"**. 2. SilverAcademy displays the Community Partner's registered facilities.

	3. The Community Partner selects a facility. 4. The Community Partner adds, updates or removes an available date and time slot. 5. SilverAcademy validates the submitted availability. 6. SilverAcademy updates the facility availability. 7. SilverAcademy displays the updated availability.
Alternative Flows:	The CP may register a new facility by selecting "Add Facility" and entering its name, address, capacity, and amenities before setting its availability.
Exceptions:	1. Ex1: If the selected slot conflicts with an approved workshop, SilverAcademy shall prevent the slot from being removed or modified. 2. Ex2: If the submitted availability is invalid, SilverAcademy shall display an error message.
Includes:	None
Special Requirements:	Facility location data must be synchronised with the OneMap API for accurate geospatial display.
Assumptions:	The CP's organisation has already been onboarded and its facilities registered in SilverAcademy.
Notes and Issues:	Consider whether recurring availability patterns (e.g. "every Tuesday evening") should be supported to reduce manual updates.

B. ApproveVenueRequests

Use Case ID:	#4-2
Use Case Name:	ApproveVenueRequests

Actor:	Community Partner
Description:	Allows a Community Partner (CP) to log, update, and manage the availability slots of the physical neighbourhood facilities (e.g. void decks, CC function rooms, badminton court), so Senior Instructors can view accurate, up-to-date venue options when requesting a workshop venue.
Preconditions:	Allows a CP to review a Senior Instructor's pending venue request for a workshop and approve or reject the booking based on facility availability and suitability.
Postconditions:	The Senior Instructor has submitted a venue request (RequestVenue) for a facility managed by the CP The requested facility and time slot fall within the CP's registered availability. The venue request status is updated to Approved or Rejected and the Senior Instructor is notified of the outcome. If approved, the requested slot is locked and marked Booked for that workshop.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Community Partner opens **"Venue Requests"**.

	2. SilverAcademy displays pending venue requests for the Community Partner's facilities. 3. The Community Partner selects a venue request. 4. SilverAcademy displays the requested venue, date, time and workshop details. 5. The Community Partner selects **"Approve"** or **"Reject"**. 6. If approved, SilverAcademy verifies that the requested slot is still available. 7. SilverAcademy updates the venue request status. 8. If approved, SilverAcademy reserves the selected venue slot. 9. SilverAcademy notifies the Senior Instructor of the decision.
Alternative Flows:	The CP may request additional information from the Senior Instructor before deciding, placing the request in an "Awaiting Info" state.
Exceptions:	1. Ex1: If rejected, SilverAcademy prompts the CP to optionally provide a reason, relayed to the Senior Instructor. 2. Ex2: If the requested slot becomes unavailable (e.g. double-booked) before a decision is made, SilverAcademy flags the request as Conflicted and prompts the CP to suggest an alternative slot.
Includes:	RequestVenue (Senior Instructor)

Special Requirements:	Pending request notifications must reach the CP within a defined SLA (e.g. within minutes of submission).
Assumptions:	The CP actively monitors their dashboard or notification channel for pending requests in a timely manner.
Notes and Issues:	Define an escalation path (e.g. to Admin) if a CP does not respond to a request within a set number of days.

VI. For Functional Requirement #5 - Admin

A. ManageAccounts

Use Case ID:	#5-1
Use Case Name:	ManageAccounts

Actor:	Admin
Description:	Allows an Admin to review and manage SilverAcademy user accounts, including viewing account information, flagging accounts, suspending accounts, and reactivating accounts when appropriate.
Preconditions:	1. The Admin is authenticated and has administrative privileges. 2. The target user account exists in SilverAcademy.
Postconditions:	The user's account status is updated accordingly, and the action is recorded in the administrative audit log. If the account is suspended, the user's access to relevant SilverAcademy functions is revoked.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Admin opens **"Manage Accounts"**. 2. SilverAcademy displays searchable user accounts. 3. Admin searches for a user. 4. SilverAcademy displays the matching account. 5. Admin selects **Flag**, **Suspend** or **Reactivate**.

	6. SilverAcademy verifies the Admin's authorisation. 7. SilverAcademy updates the account status. 8. SilverAcademy records the action in the audit log.
Alternative Flows:	If the Admin selects Flag, the account remains active but is marked for administrative monitoring. If the Admin selects Reactivate, SilverAcademy restores the user's account access after confirmation.
Exceptions:	Ex1: If the selected account does not exist or has already been modified by another Admin, SilverAcademy displays an error and refreshes the account information. Ex2: If the Admin does not have sufficient privileges for the requested action, SilverAcademy denies the operation and records the attempted access.
Includes:	None
Special Requirements:	Role-based access control must restrict account management functions to authorised Admins. Administrative actions involving account status changes must be logged for auditing purposes.
Assumptions:	Admins are authorised to manage user accounts according to SilverAcademy platform policies.
Notes and Issues:	Suspension rules and whether certain accounts require secondary Admin approval should be defined.

B. ManageWorkshops

Use Case ID:	#5-2
Use Case Name:	ManageWorkshops

Actor:	Admin
Description:	Allows an Admin to review and manage published workshop listings, including editing workshop information, archiving workshops, overriding workshop status when necessary, and maintaining workshop category classifications.
Preconditions:	1. The Admin is authenticated and has administrative privileges. 2. The target workshop exists in SilverAcademy.
Postconditions:	The workshop information or status is updated, archived, or overridden as requested. The change is reflected in the workshop directory and recorded in the administrative audit log.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Admin opens **"Manage Workshops"**. 2. SilverAcademy displays the workshop directory. 3. Admin searches for and selects a workshop. 4. SilverAcademy displays the workshop details. 5. Admin selects an authorised management action. 6. SilverAcademy validates the requested change.

	7. SilverAcademy applies the change. 8. SilverAcademy records the administrative action.
Alternative Flows:	If the Admin edits the workshop category or description, SilverAcademy saves the revised information after validation. If the Admin archives a workshop, SilverAcademy removes it from public discovery while preserving historical booking and attendance records.
Exceptions:	Ex1: If the workshop has active bookings and the requested administrative action would invalidate existing bookings, SilverAcademy displays a conflict warning and requires the Admin to confirm the impact or initiate the appropriate cancellation/rescheduling process. Ex2: If the workshop has already started or completed, SilverAcademy restricts changes to fields that would compromise historical records.
Includes:	None
Special Requirements:	Workshop status transitions must follow predefined business rules. Historical booking and attendance records must not be deleted when a workshop is archived.
Assumptions:	Admins are authorised to override workshop information when required for safety, policy, or operational reasons.

Notes and Issues:	A controlled category taxonomy should be maintained to ensure consistent workshop discovery and filtering.

C. ResolveUserReports

Use Case ID:	#5-3
Use Case Name:	ResolveUserReports

Actor:	Admin
Description:	Allows an Admin to review, investigate, and resolve incident reports submitted by Learners or Volunteers, including reports concerning user conduct, venue safety, workshop issues, or software problems.
Preconditions:	1. The Admin is authenticated and has administrative privileges. 2. At least one incident report exists in the Admin Incident Queue.
Postconditions:	The incident report is updated with its latest status and resolution details. Any required administrative action, such as suspending a user or cancelling a workshop, is executed. The resolution and relevant actions are recorded in the audit log.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Admin opens the **Admin Incident Queue**. 2. SilverAcademy displays submitted incident reports. 3. Admin selects an incident report. 4. SilverAcademy displays the report details and supporting evidence. 5. Admin reviews the incident.

	6. Admin changes the incident status to an appropriate status. 7. If additional information is required, Admin sends a request for information to the relevant user. 8. Admin records the investigation outcome. 9. If disciplinary action is required, Admin performs an authorised administrative action. 10. Admin records the incident resolution. 11. SilverAcademy notifies the relevant user of the updated incident status or resolution. 12. SilverAcademy records all administrative actions in the audit log.
Alternative Flows:	If additional information is required, the Admin changes the incident status to Awaiting Information and sends a message to the relevant user. If the report is determined to be invalid or unfounded, the Admin records the reason and closes the report without disciplinary action. If the issue is a software defect, the Admin may classify the report as a technical issue for follow-up by the development team.
Exceptions:	1. Ex1: If the report contains insufficient information, Admin shall be able to request additional information. 2. Ex2: If supporting evidence cannot be accessed, SilverAcademy shall display an appropriate error message.

	3. Ex3: If an Admin attempts an action outside their authorisation, SilverAcademy shall reject the action. 4. Ex4: If the incident has already been resolved, SilverAcademy shall display its current status and prevent conflicting updates.
Includes:	SubmitUserReport
Special Requirements:	SilverAcademy shall restrict incident reports and supporting evidence to authorised users and shall record all administrative actions with the Admin ID, timestamp and action performed.
Assumptions:	Admins have access to the information necessary to investigate incidents and are authorised to perform disciplinary actions.
Notes and Issues:	Incident severity levels and response-time targets should be defined. Critical safety incidents should be prioritised over general feedback or software issues.

D. ManageVoucherInventory

Use Case ID:	#5-4
Use Case Name:	ManageVoucherInventory

Actor:	Admin
Description:	Allows an Admin to upload, validate, monitor, and manage pre-purchased supermarket voucher codes in SilverAcademy's internal voucher inventory.
Preconditions:	1. The Admin is authenticated and has administrative privileges. 2. The Admin has valid pre-purchased voucher codes available for upload.
Postconditions:	Valid voucher codes are securely stored in the voucher inventory and made available for eligible Senior Instructors to redeem. Invalid or duplicate codes are rejected and are not added to the available inventory.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. Admin opens **"Voucher Inventory"**. 2. Admin uploads a voucher code file. 3. SilverAcademy validates the uploaded voucher codes. 4. SilverAcademy rejects duplicate or invalid voucher codes. 5. SilverAcademy stores valid voucher codes as **Unclaimed**.

	6. SilverAcademy displays the updated voucher inventory. 7. Admin can view the current voucher inventory status.
Alternative Flows:	The Admin may filter the inventory by voucher provider, denomination, or status. The Admin may mark an unused voucher as Invalid or Removed when required by operational policy.
Exceptions:	1. Ex1: If the uploaded file contains duplicate voucher codes, SilverAcademy shall reject the duplicate codes. 2. Ex2: If a voucher code has an invalid format, SilverAcademy shall reject the code. 3. Ex3: If the upload fails, SilverAcademy shall not partially update the voucher inventory.
Includes:	None
Special Requirements:	Voucher uploads shall be processed as a transaction so that incomplete uploads do not create inconsistent inventory records.
Assumptions:	Admins obtain voucher codes through legitimate pre-purchased voucher arrangements.
Notes and Issues:	The system should maintain an audit trail for voucher uploads, status changes, and claims without exposing voucher codes unnecessarily.

E. ReviewInstructorAndCurriculum

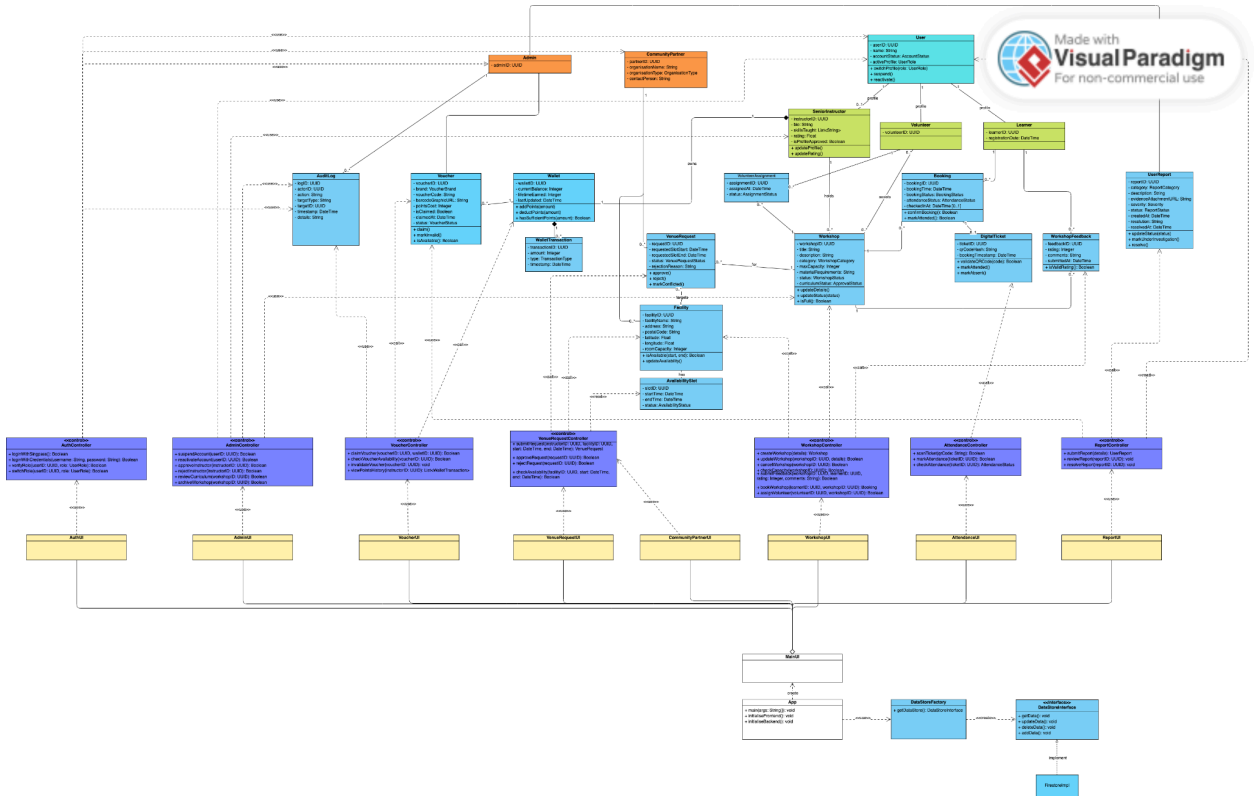
Use Case ID:	#5-5
Use Case Name:	ReviewInstructorAndCurriculum

Actor:	Admin
Description:	Allows an Admin to review and approve or reject pending Senior Instructor profiles and workshop curriculum to ensure that instructors and proposed activities satisfy SilverAcademy's safety and platform requirements before workshops are made available for public registration.
Preconditions:	1. The Admin is authenticated and has administrative privileges. 2. A Senior Instructor profile or workshop curriculum is pending review. 3. The required profile and curriculum information has been submitted.
Postconditions:	The reviewed profile or curriculum is marked as Approved, Rejected, or Requires Revision. Approved Senior Instructors may access instructor functions, while approved curriculum may proceed to the appropriate workshop and venue approval stages.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. Admin opens **"Instructor and Curriculum Review"**.

	2. SilverAcademy displays pending Senior Instructor profiles and curriculum submissions. 3. Admin selects a submission. 4. SilverAcademy displays the instructor profile and curriculum details. 5. Admin reviews the submitted information. 6. Admin selects **Approve**, **Reject** or **Request Revision**. 7. SilverAcademy updates the submission status. 8. SilverAcademy records the Admin's decision. 9. SilverAcademy notifies the Senior Instructor of the decision.
Alternative Flows:	If revisions are required, SilverAcademy changes the submission status to Requires Revision and allows the Senior Instructor to update and resubmit the information. If the Admin approves the submission, SilverAcademy enables the corresponding instructor or curriculum functions.
Exceptions:	1. Ex1: If required information is missing, Admin shall be able to request revision. 2. Ex2: If the submission is rejected, SilverAcademy shall record the rejection status. 3. Ex3: If an unauthorized Admin attempts to approve or reject a submission, SilverAcademy shall reject the action.
Includes:	None

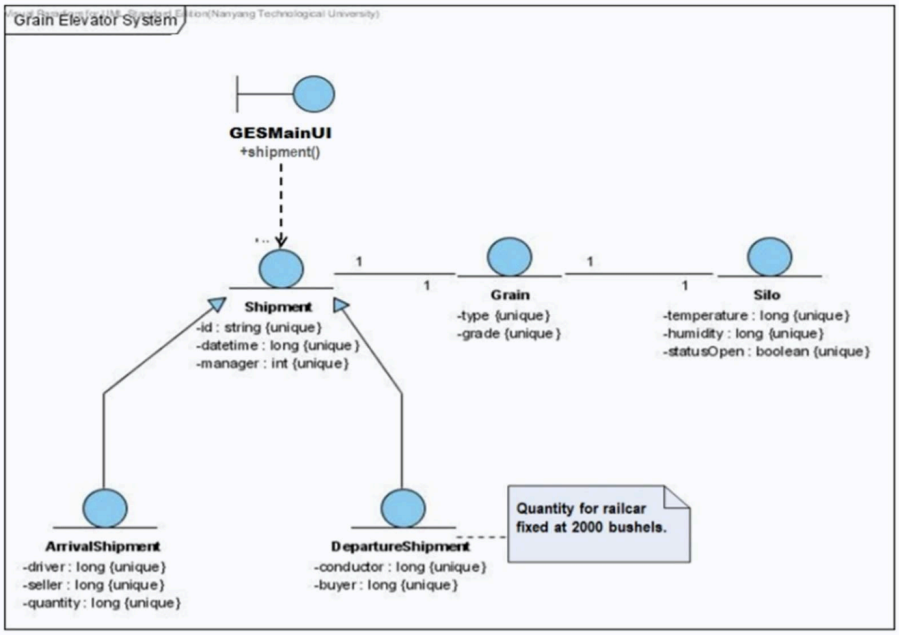
Special Requirements:	Approval decisions must be recorded with the reviewing Admin, timestamp, decision, and reason where applicable. Only approved Senior Instructors and curriculum should proceed to the relevant subsequent workflow stages.
Assumptions:	Admins have defined safety and eligibility criteria against which submissions can be evaluated.
Notes and Issues:	The project should define whether instructor approval and curriculum approval are independent states or whether both must be approved before a workshop can proceed.

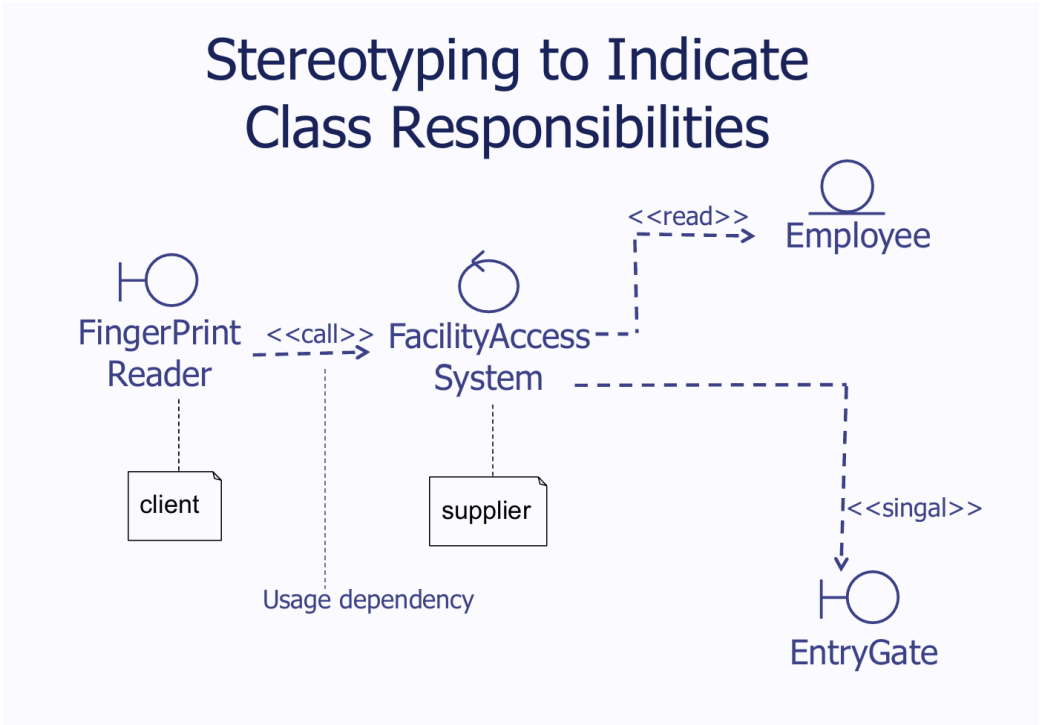
5. Class Diagram of Entities



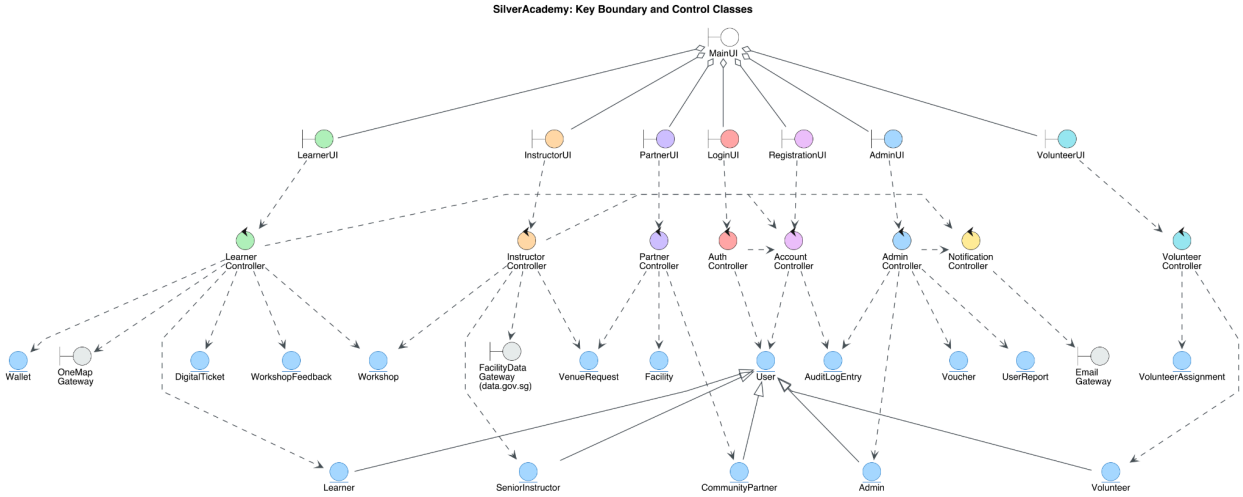
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Question 1: Grain Elevator System (GES)
(b) Initial class diagram



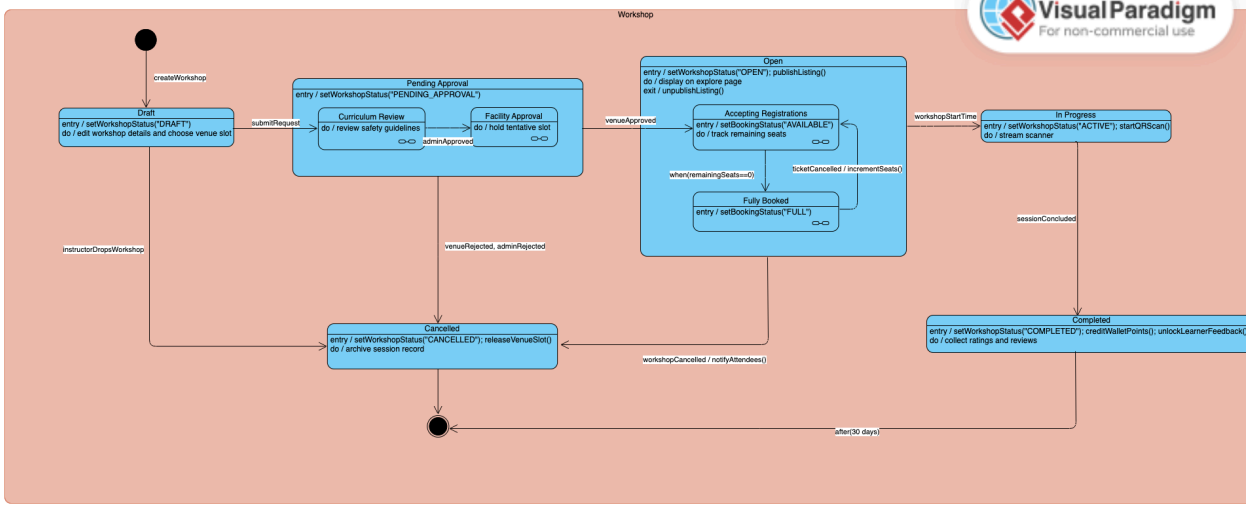


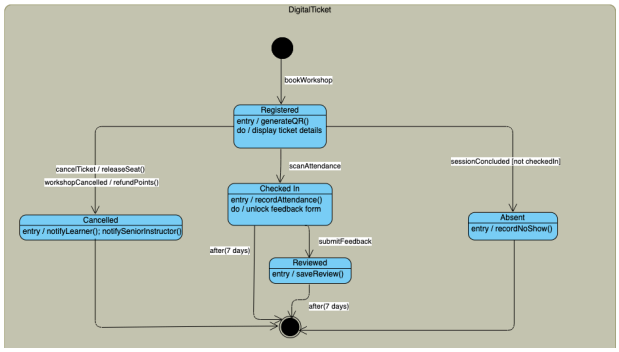
6. Key Boundary Classes and Control Classes

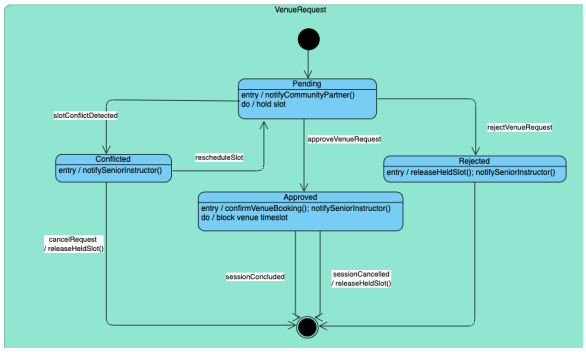


- Boundary class icon (circle with a “|-” on left): a boundary class. These are screens the user interacts with, such as LoginUI or LearnerUI, or gateways to external systems, like OneMap Gateway or Email Gateway.
- Control class icon (circle with a curved arrow on top): a control class. There is one per actor role, for example LearnerController or AdminController. Each holds the use-case logic for that role.
- Entity class icon (circle with line below): an entity class. These hold persistent data, like User, Workshop or Wallet. Their attributes and relationships are shown in the Entity Class Diagram.
- Dashed arrows: "use" dependencies. The class at the tail calls or depends on the class at the head, for example LearnerUI uses LearnerController, and LearnerController uses Workshop.
- Solid lines with a hollow diamond at MainUI: aggregation. MainUI is the application shell that hosts each role's screen.
- Solid lines with a hollow triangle at User: inheritance. Learner, SeniorInstructor, Volunteer, CommunityPartner and Admin are subclasses of User.

7. Dialog Map



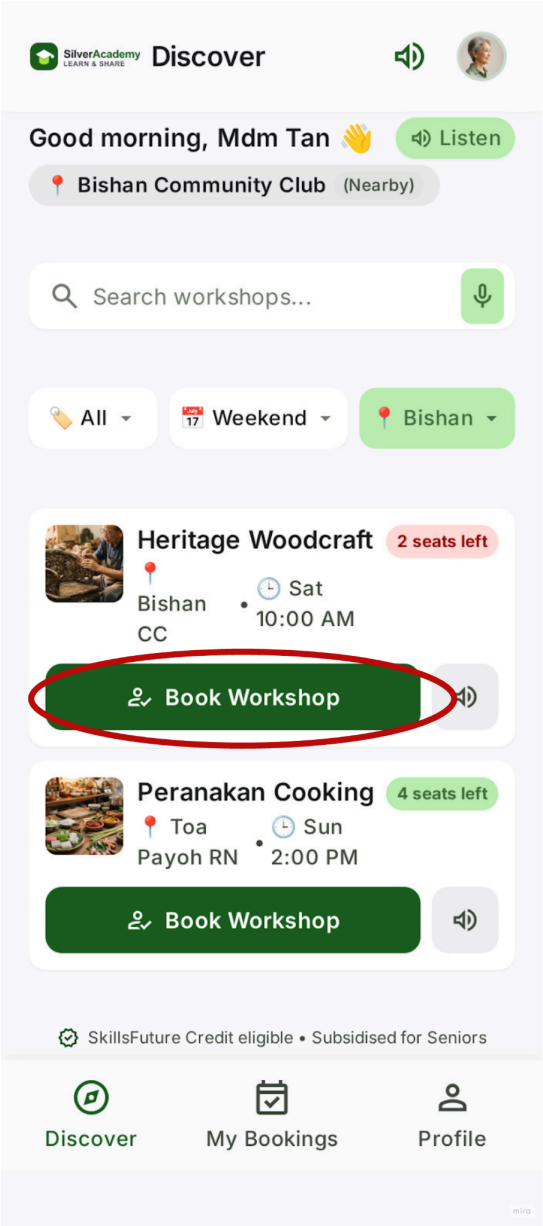




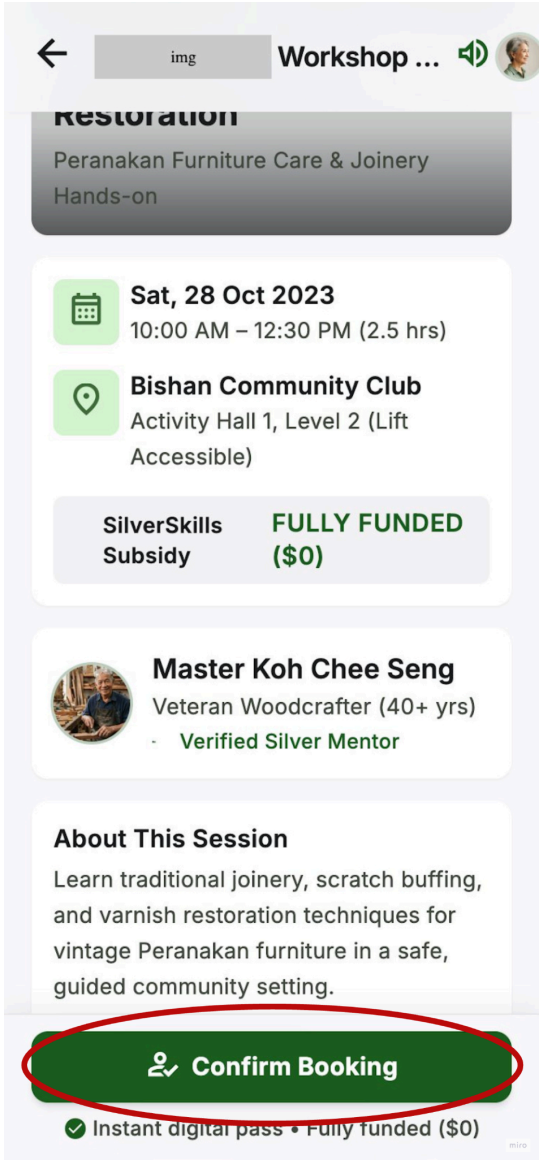
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8. UI Mockup

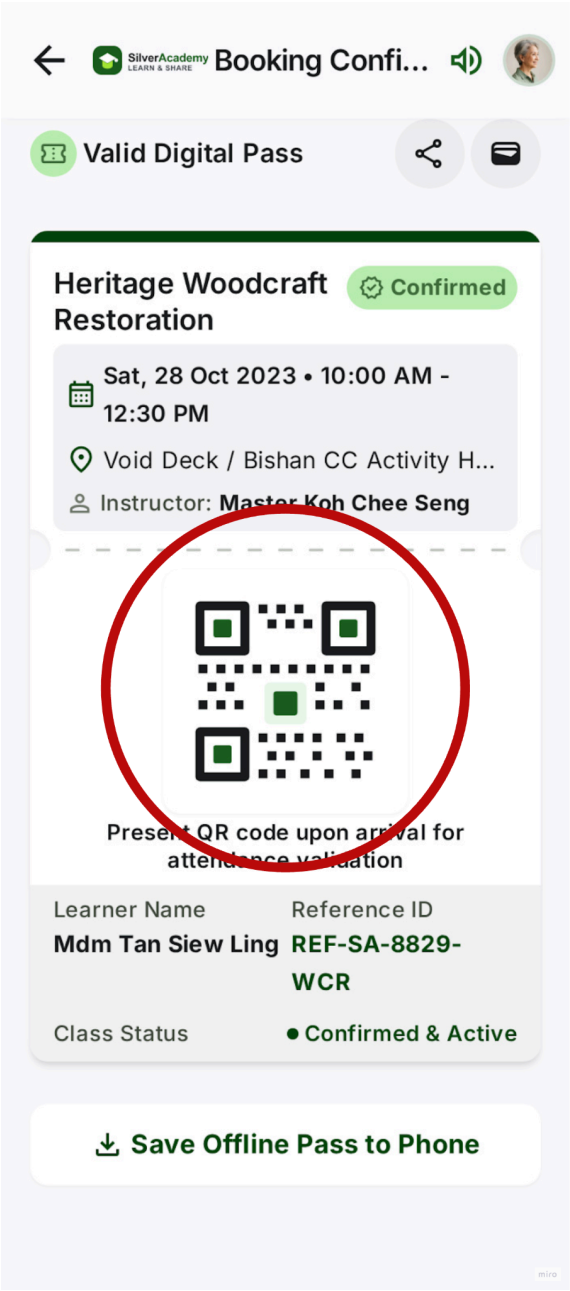
1. Learner Booking Journey



Step 1: The Learner browses the Discover feed, views available seats, and taps the "Book Workshop" button under their desired class to view more details.

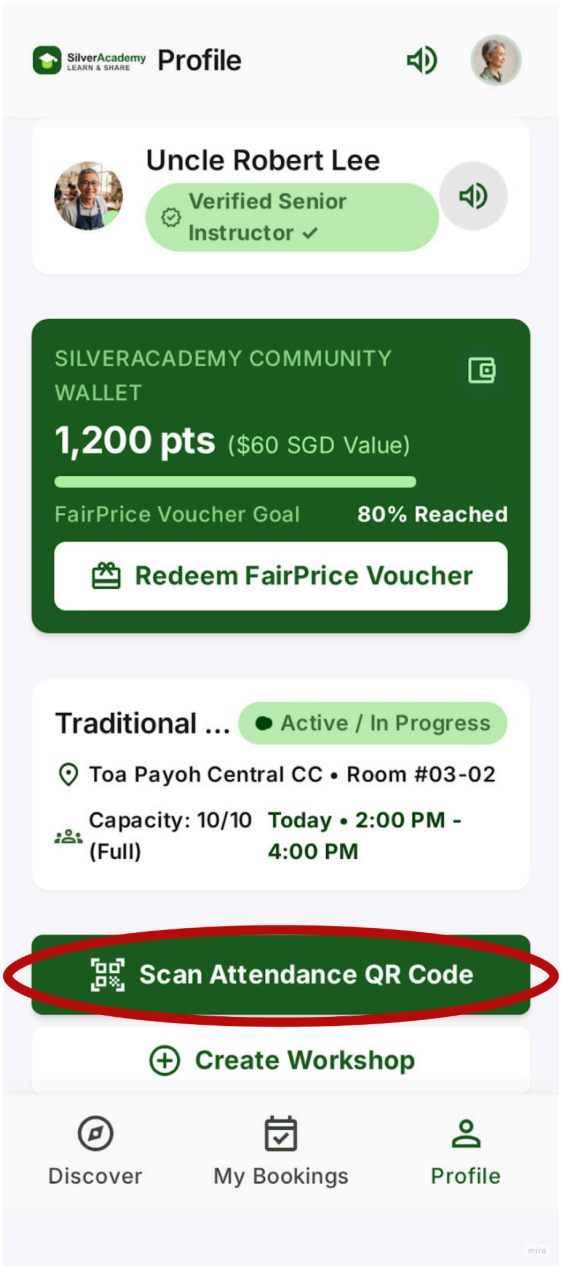


Step 2: The Learner reviews the workshop information, including the venue and instructor profile, and taps the "Confirm Booking" button at the bottom of the screen.

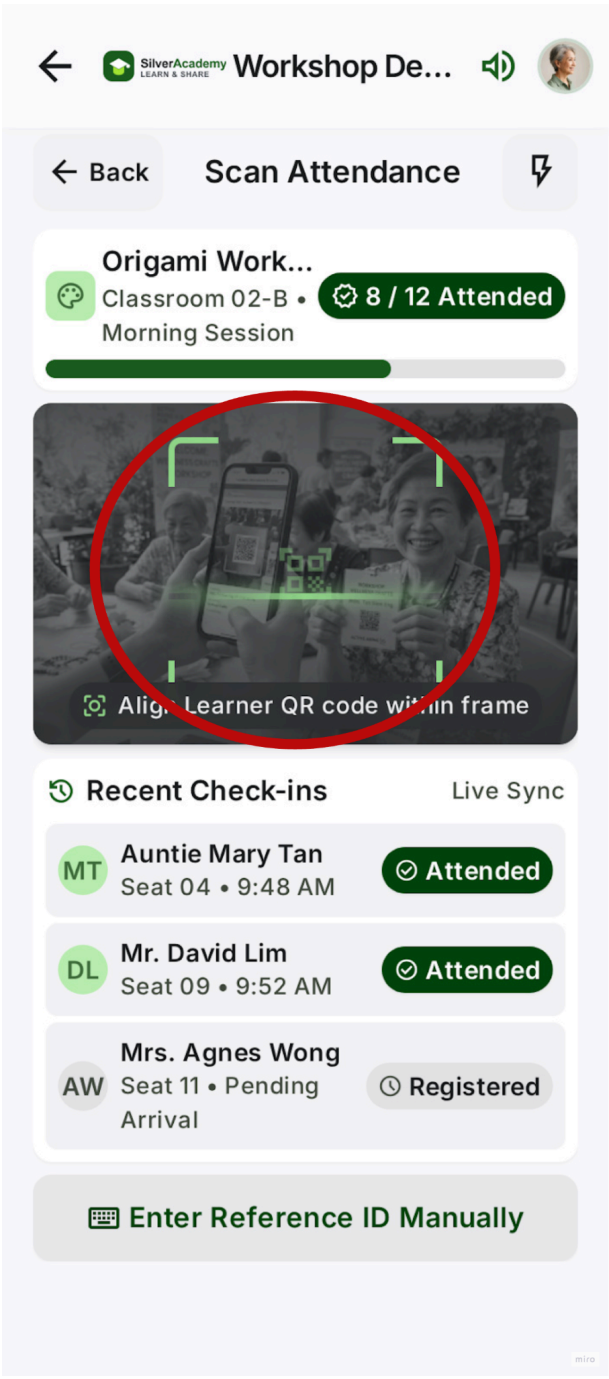


Step 3: The system processes the registration and issues a digital pass with a unique QR code for the Learner to present upon arrival at the venue.

2. Senior Instructor Check-In Flow



Step 1: The Senior Instructor reviews their active sessions from the profile dashboard and taps the "Scan Attendance QR Code" button to initiate the check-in process.



Step 2: The device camera activates, allowing the instructor to scan the Learner's digital ticket, which automatically updates the live-sync attendance roster below.

Technology Stack Discussion

For an entry-level team, I'd optimize for **few technologies, minimal setup, and a mostly monolithic architecture** rather than a "production-scale" stack.

Area	Suggested technology	Why
Language	**TypeScript**	Same language everywhere
Mobile app	**React Native + Expo**	Easy cross-platform app; camera/QR support
Admin/Partner web app	**React**	Simple and shares React knowledge with mobile
Backend	**Firebase Cloud Functions**	No server setup; backend logic in one place
Database	**Firebase Firestore**	Very quick to set up; no SQL/database server management
Authentication	**Firebase Auth + Singpass integration**	Firebase handles sessions/users; Singpass used where required
File storage	**Firebase Storage**	Easy storage for report attachments, etc.
Maps	**OneMap API**	Already required by SilverAcademy
QR	**Expo Camera + QR library**	Required for attendance scanning
Hosting	**Firebase Hosting/Vercel**	Deploy web frontend with little configuration

React Native app + React web app → Firebase → Firestore

Keep all backend functionality: bookings, workshops, attendance, vouchers, users in one Firebase project, organised into modules rather than separate microservices. This matches the system's required functional separation while staying easy to build and maintain.

For this project, I would choose Firebase over PostgreSQL/NestJS mainly because it removes a lot of database, deployment, authentication, and server configuration work.